

Total number of printed pages-7

1-23 (MAT1104C)

2023

MATHEMATICS

(Core Course)

Paper : MAT1104C

(**Classical Algebra**)

Full Marks : 60

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $2 \times 5 = 10$

(a) Find all the complex numbers such that

✓ $z^3 = \bar{z}$.

(b) Is De Moivre's theorem valid for irrational numbers ? Justify.

✓ (c) Determine the nature of the roots of the equation $x^7 + x^5 - x^3 = 0$.

✓ (d) Define basic and free variables of a system of linear equations.

1 0 0 Contd.
- - -

- (e) If α, β, γ are roots of the equation $x^3 + \frac{3A}{a_0^2}x + \frac{3}{a_0^3} = 0$, find the equation whose roots are $a_0\alpha, a_0\beta, a_0\gamma$.

2. Answer **either** (a) **or** (b) :

- ✓ (a) Let $\varepsilon_0, \varepsilon_1, \varepsilon_2, \dots, \varepsilon_{n-1}$ be the n th roots of unity. For any positive integers k , prove that

$$\sum_{j=0}^{n-1} \varepsilon_j^k = \begin{cases} n & \text{if } n/k \\ 0, & \text{otherwise} \end{cases}$$

- (b) If $x = \log \tan \left(\frac{\pi}{4} + \frac{y}{2} \right)$, show that

$$y = -i \log \tan \left(\frac{ix}{2} + \frac{\pi}{4} \right).$$

3. Answer **either** (a) **or** (b) :

- ✓ (a) Prove that the roots of the equation

$$\frac{1}{x-1} + \frac{2}{x-2} + \frac{3}{x-3} = x \text{ are all real.}$$

- (b) If α, β, γ are roots of the equation $x^3 + bx + c = 0$, find the value of

(i) $\sum \alpha^5$

(ii) $\sum \frac{1}{\alpha}$

$$-10 + 18$$

4. Answer **either** (a) **or** (b) :

- ✓ (a) Reduce the following matrix in reduced echelon form and hence find its rank

$$\begin{pmatrix} 0 & 3 & -6 & 6 & -5 \\ 3 & -7 & 8 & -5 & 9 \\ 3 & -9 & 12 & -9 & 15 \end{pmatrix}$$

- (b) Let us consider the matrix

$$A = \begin{pmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{pmatrix} \text{ and } b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

How many rows of A contain a pivot position? Does the equation $Ax = b$ have a solution for each b ?

5. Answer **either** (a) or (b) :

(a) Solve the equation

$$16x^4 - 64x^3 + 56x^2 + 16x - 15 = 0$$

whose roots are in A.P.

(b) Find all the complex numbers Z such that $|Z|=1$ and $\left|\frac{Z}{\bar{Z}} + \frac{\bar{Z}}{Z}\right| = 1$.

6. Answer **either** (a) or [(b) and (c)] : 10

(a) Prove that the primitive roots of

$$Z^m - 1 = 0 \text{ are } \varepsilon_k = \cos \frac{2k\pi}{m} + i \sin \frac{2k\pi}{m}$$

where $0 \leq k < m$ and $\gcd(k, m) = 1$.

Hence find the primitive roots of the equation $Z^{16} - 1 = 0$. 6+4=10

(b) Let $P_0, P_1, P_2, \dots, P_{n-1}$ be a regular polygon inscribed in a circle of radius 1. Prove that

$$\sin \frac{\pi}{n} \sin \frac{2\pi}{n} \dots \sin \frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$$

5

$$|z| = 1$$

5

3

(c)

Show that

$$\tan^{-1}(\cot \theta \tanh \phi) = \frac{1}{2i} \log \left[\frac{\sin(\theta + i\phi)}{\sin(\theta - i\phi)} \right]$$

5

7. Answer **either** (a) or (b) :

10

(a) Describe the Cardan's method to solve the cubic equation

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0; a_0 \neq 0.$$

Hence solve the equation

$$x^3 - 15x^2 - 33x + 847 = 0$$

5+5=10

(b) Find the equation whose roots are squares of the differences of the roots of the cubic equation

$$a_0x^3 + 3a_1x^2 + 3a_2x + a_3 = 0.$$

Discuss the nature of roots for

$$G^2 + 4H^3 < 0 \text{ and } H < 0, \text{ where}$$

$$H = a_0a_2 - a_1^2$$

$$G = a_0^2a_3 - 3a_0a_1a_2 + 2a_1^3$$

8+2=10

$$\begin{array}{r} 681 \\ 9 \overline{) 681} \\ \underline{54} \\ 141 \\ \underline{126} \\ 15 \end{array}$$

$$\begin{array}{r} 2671 \\ 1 \overline{) 2671} \\ \underline{2000} \\ 671 \\ \underline{600} \\ 71 \end{array}$$

8. Answer **either** [(a) and (b)] **or** (c) : 10
- (a) Describe the solution of the following system in parametric vector form and provide a geometrical comparison with the solution of the homogeneous system: 5

$$\begin{aligned}x_1 + 3x_2 - 5x_3 &= 4 \\x_1 + 4x_2 - 8x_3 &= 7 \\-3x_1 - 7x_2 + 9x_3 &= -6\end{aligned}$$

(b) Let $A = \begin{pmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{pmatrix}$, $u = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $b = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix}$

$c = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$ and define the transformation

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(x) = Ax$

(i) Find $T(u)$

(ii) Find an x in $\mathbb{R}^2$ whose image under T is b .

$2+3=5$

- (c) Describe Cramer's rule to solve the following system of equations :

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 &= b_1 \\a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 &= b_2 \\a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 &= b_3 \\a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 &= b_4\end{aligned}$$

Hence solve by Cramer's rule

$$\begin{aligned}4x_1 + x_2 + x_4 &= 2 \\x_1 + 4x_2 + x_3 &= -2 \\x_2 + 4x_3 + x_4 &= 2 \\x_1 + x_3 + 4x_4 &= -2\end{aligned}$$

$5+5=10$